

Software Requirements Specification (SRS)

Project: Sales & Incentive Automation Tool (SIAT)

Version: 1.0

Status: Final Draft

1. Introduction

The Sales & Incentive Automation Tool (SIAT) is a Python-based application designed to automate the reconciliation of mobile phone sales data. The system transforms raw transaction logs into finalized financial reports by calculating price protections, promotional schemes, and Net Landing Costs (NLC).

2. System Overview

The system follows an ETL (Extract, Transform, Load) architecture. It ingests data from four primary Excel/CSV sources and outputs a formatted master report including a dashboard for management analytics.

3. Functional Requirements

3.1 Data Ingestion

The system must handle files with irregular structures, including:

- Headers located on Row 2 (index 1).
- Files with no headers (specifically the Drop Dump).
- Encoding support for Latin-1 and UTF-8 to handle special characters.

3.2 Core Calculation Engine (11-Step Logic)

Step	Requirement / Formula
Step 1-2	Drop Detection: Match IMEI in 'Drop Dump'. If match found, populate Column K.
Step 3-4	Price Match: Lookup Master Model in 'Price List'. Validation: Sell Out Date must be between 'Valid From' and 'Valid To'.
Step 5-6	Tax Base: Final Price = Purchase Price - Drop. Pre-GST Price = Final Price / 1.18.
Step 7-8	Scheme Application: Match Master Model in 'Scheme' file. Apply Percentage (%) to Pre-GST price or apply Flat Payouts based on Sell Out Date.
Step 9	Incentive Sum: Aggregate all Pct and Flat schemes into 'Total Incentive Received'.
Step 10	Net Landing Cost (NLC): Final Price - Total Incentive.
Step 11	Pivot Reporting: Generate Distributor-wise totals for all incentive categories.

4. User Interface & Dashboard Requirements

The application must include a professional, 'eye-catchy' dashboard (Web-based using Streamlit/Plotly or modern GUI). Essential visualizations include:

- • Total Incentive Recovery: Owed by Distributor.
- Margin Analysis: Purchase Price vs NLC.
- Payout Trends: Growth/decline of incentives month-over-month.
- Error Logs: Visual alerts for missing models or duplicate IMEIs.

5. Non-Functional Requirements

- • Responsive Design: Dashboard must scale for desktop and tablet views.
- Scalability: Must handle thousands of rows of data across multiple brands.
- Portability: Deliverable as a one-click executable (.exe) or Python script.

6. Data Integrity & Validations

- Duplicate Check: Prevent multiple payouts for the same IMEI.
- Fuzzy Matching: Handle minor model name variations (e.g., 'C71 4@64' vs 'C71 4/64').
- Date Validation: Flag sales where the Sell Out date is earlier than the Purchase date.